Classifying Eye State from EEG Signals via k-Nearest Neighbours: A Cross-Validation Honesty Study

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Abstract

We pick a single data-analysis approach from the COGS 109 study guide—**classification**—and a single model within that approach—**k-nearest neighbours (KNN)**—and we perform model selection by sweeping the hyperparameter k over a log-spaced grid $\{1, 3, 5, 7, 11, 15, 21, 31, 51, 75, 99, 151, 201\}$ to find the value that minimises 5-fold cross-validation (CV) error on the UCI #264 EEG Eye State dataset (Roesler, 2013). We then run that *same* model-selection procedure under three different cross-validation schemes: shuffled, naive blocked, and stratified blocked. The contribution is *methodological*: we document how the choice of CV scheme changes the selected hyperparameter and the reported accuracy on an autocorrelated EEG recording. All three schemes select $k = 1$, but the reported accuracies differ dramatically: $97.3\% \pm 0.4\%$ under shuffled CV, $50.0\% \pm 10.7\%$ under naive blocked CV, and $77.8\% \pm 2.7\%$ under honest stratified blocked CV. The $19.5\,pp$ gap between the shuffled and stratified-blocked estimates is leakage attributable to scheme choice—not to model quality, since the identical KNN model is evaluated in all three settings on the same training data. A sanity-check appendix runs the same idea on three alternative classifiers (LDA, PCA→LDA, PCR-as-classifier) and finds leakage gaps of only 2–6 pp, consistent with KNN being uniquely vulnerable to lag-1 autocorrelation because it predicts from per-sample similarity.

Shuffled CV says we have a 97% classifier. Honest stratified blocked CV says we have a 78% classifier. The difference is leakage, not signal.

1 Introduction

1.1 Problem and motivation

Electroencephalography (EEG) eye-state classification—deciding from raw scalp voltages whether a subject's eyes are open or closed—is a building block of brain–computer interfaces, drowsiness detection, and attention-monitoring systems. The UCI #264 EEG Eye State dataset is one of the most widely used benchmarks for this task. The dataset's originators report a best error of 2.7% ($\approx$ 97.3% accuracy) using the instance-based KStar classifier under Weka's default (shuffled) 10-fold cross-validation, while non-instance-based methods score far worse (Roesler & Suendermann, 2013); online tutorials and student write-ups likewise routinely report k-nearest-neighbour accuracies in the 95–98% range under standard shuffled k-fold cross-validation.

We argue in this report that those numbers reflect a *methodological artefact* rather than a strong classifier. The dataset is a single ∼117-second continuous recording from one subject; its label changes only 23 times across 14,980 samples; adjacent voltage samples are nearly identical (per-channel lag-1 autocorrelation ≈ 0.97); and the label is piecewise-constant over runs that are sometimes thousands of samples long (lag-1 *label* autocorrelation ≈ 0.997). A KNN classifier with $k = 1$ evaluated under shuffled

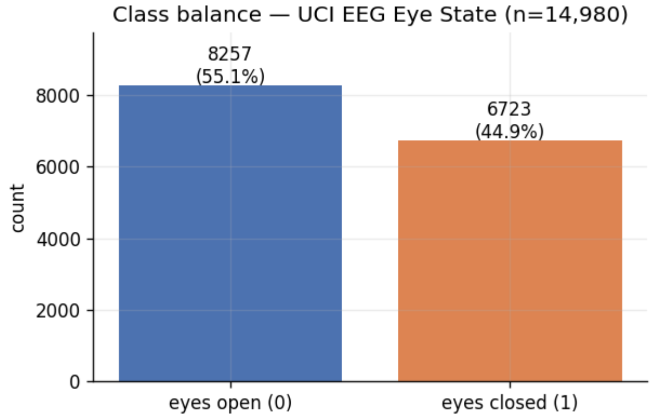

Figure 1: Class balance of the UCI #264 EEG eye-state dataset. The majority class (eyes open) sets the random-guess baseline at 0.5512.

k-fold CV is essentially asking "what was the eye state of the sample 8 ms before or after this one?"—a question to which it can almost always answer correctly. That is a property of the cross-validation rule, not of the classifier.

The question this report addresses is therefore methodological rather than predictive:

> *Given an autocorrelated single-subject EEG recording and a* KNN *classifier, how does the choice of cross-validation scheme change the result of an otherwise identical model-selection procedure?*

This matters because cross-validation is the standard tool students are taught for honest accuracy estimation, yet on autocorrelated time-series data the default i.i.d. shuffled scheme silently inflates the estimate. Quantifying that inflation on a clean, well-known benchmark makes the failure mode concrete and reproducible.

1.2 Dataset and key characteristics

We use the UCI EEG Eye State dataset (Roesler, 2013; UCI Machine Learning Repository #264): a single uninterrupted recording from one subject wearing an Emotiv EPOC consumer-grade EEG headset.

- **Observations:** $n = 14{,}980$ samples, recorded at $\sim$128 Hz over $\sim$117 s.
- **Predictors:** $p = 14$ continuous voltage channels (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4).
- **Response:** the binary label `eyeDetection` (0 = eyes open, 1 = eyes closed), annotated by the experimenter.
- **Class balance:** 55.12% open (8,257 samples) vs. 44.88% closed (6,723 samples), so the majority-class baseline—the accuracy floor any useful classifier must beat—is 0.5512 (Figure 1).
- **Noise sources:** consumer-grade hardware prone to electrode drift, motion and muscle artefacts, and occasional headset disconnections (visible as a handful of extreme-voltage outliers).

A purely structural feature of this dataset is the heart of the study. The labels change rarely: the recording contains only **24 contiguous label runs** (the subject opens or closes their eyes only 23 times in $\sim$117 s), with run lengths ranging from under a hundred to several thousand consecutive samples (Figure 2). Consequently the lag-1 autocorrelation of a single voltage channel is high (≈ 0.97 on average, ranging 0.91–0.99 across the 14 channels after outlier removal)—adjacent samples are nearly identical—and the label, being piecewise-constant over long runs, is even more persistent: its autocorrelation is ≈ 0.997 at lag 1, ≈ 0.83 at lag 50 (≈ 0.4 s), crossing zero only at very long lags (Figure 3).

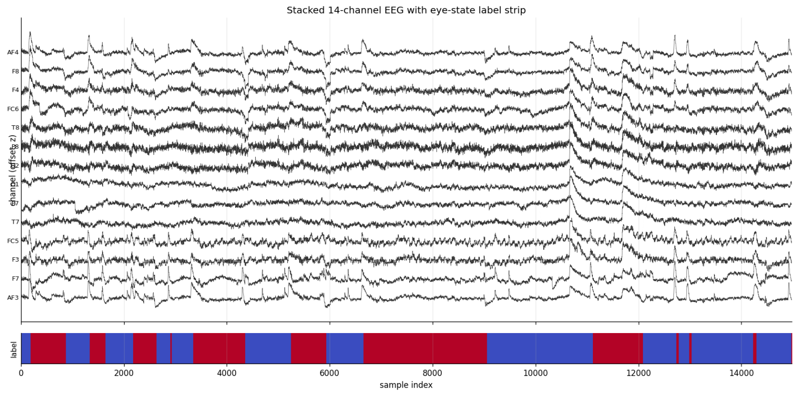

Figure 2: Raw 14-channel EEG voltages (top) with the eye-state label strip (bottom; blue = eyes open, 0; red = eyes closed, 1). Channels are continuous and smooth on the millisecond scale; the label is piecewise-constant over multi-second runs.

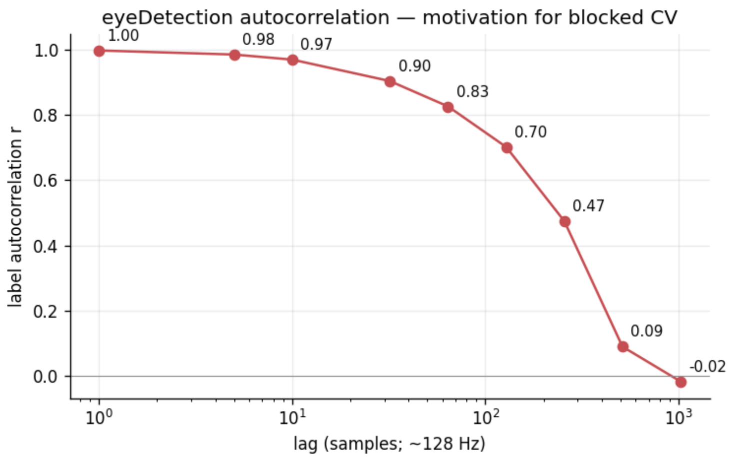

Figure 3: Autocorrelation of the binary `eyeDetection` label across lags 1–1000. The label is essentially identical to itself for the first ∼100 samples (∼0.8 s) and only decorrelates at very long lags. This is the structural reason shuffled CV leaks: a sample's nearest neighbour in time is almost always in its own class.

This is the structural problem at the heart of the report: *any* CV protocol that splits samples uniformly at random leaves each test sample with a near-duplicate in its own training set. A model whose prediction depends on per-sample similarity—KNN with $k = 1$ being the canonical example—will appear spectacular, but the score reflects memorisation, not a generalisable voltage-to-label rule.

1.3 Hypothesis

> **Hypothesis.** A KNN classifier trained on raw 14-channel EEG voltages can predict per-sample eye state above the 55.12% majority-class baseline, but accuracy estimates obtained under shuffled k-fold cross-validation will substantially exceed honest out-of-time estimates because of temporal leakage between adjacent samples.

The hypothesis has two parts and we test both: (i) the honest accuracy beats the majority baseline, and (ii) the shuffled estimate is inflated relative to a leakage-resistant estimate. A negative or ambiguous result on either part would still be informative; we report exactly what we find.

2 Methods

2.1 Approach: classification (prediction)

The COGS 109 study guide (Mukamel, 2026) enumerates several broad approaches (linear, polynomial and spline regression, PCR, LDA, KNN, K-means, hierarchical clustering, PCA). The target variable here is the binary `eyeDetection` label, so **classification** is the natural approach: we want a model that maps a 14-dimensional voltage vector to one of two discrete labels. Regression approaches do not target a discrete label without artificial thresholding, and clustering approaches ignore the label entirely. We are focused on **prediction** (out-of-sample accuracy of the eye-state label), not on inference about which channel "causes" an eye state; the methodological question is specifically about how honestly we can *estimate* that predictive accuracy.

2.2 Model: k-nearest neighbours

Within classification we chose KNN as our model (implemented with scikit-learn; Pedregosa et al., 2011). KNN assigns a test sample the majority label among its k nearest training samples under Euclidean distance in the z-scored 14-D channel space. We chose it for three reasons. First, it is the simplest non-parametric classifier in the palette: it makes no distributional assumption and has exactly one hyperparameter (k) controlling the bias–variance trade-off (James et al., 2021), making it ideal for a clean single-axis hyperparameter sweep. Second—and most importantly for this study—because KNN predicts from per-sample similarity to the training set, it is the model in the palette that *most directly* exploits the lag-1 autocorrelation of EEG data; if we want to study how CV scheme choice changes a result, picking the model most sensitive to that choice gives the cleanest empirical signal. Third, KNN at small k reproduces the ∼97% literature numbers on this dataset under shuffled CV, so the leakage gap we measure is a like-for-like comparison between an honest and a leaky estimate of the same baseline.

2.3 Preprocessing

1. **Outlier removal.** We drop the four samples with at least one channel exceeding 4 standard deviations above the channel mean ($z > 4$). This is standard EEG hygiene for headset disconnections and artefacts; an ablation (`tables/02_outlier_ablation.csv`) shows it does not materially change downstream accuracy but stabilises per-channel summary statistics. Cleaning leaves 14,976 samples.

2. **Chronological split.** We split the cleaned recording chronologically into 80% train ($n = 11{,}917$) and 20% test ($n = 2{,}995$), inserting a **64-sample seam gap** (0.5 s at 128 Hz) between train and test so no train sample is within half a second of a test sample. This is the minimum cross-partition isolation warranted by the autocorrelation profile in Figure 3.

3. **z-scoring (train statistics only).** We z-score each channel using the *training* mean and standard deviation only, and reuse those parameters for the test set. z-scoring before the split would itself leak (the test set would shape the channel scale), so we defer it to after the split.

2.4 Model-selection procedure: the k-sweep

The hyperparameter to tune for KNN is k, the number of neighbours that vote. We sweep

$$k \in \{1, 3, 5, 7, 11, 15, 21, 31, 51, 75, 99, 151, 201\},$$

a log-spaced, odd-only grid (odd avoids ties on a binary label) that starts at $k = 1$ (the most leakage-prone setting) and extends to $k = 201$ (a local average that should damp out lag-1 similarity if a real classification signal exists). For each k and each CV scheme we run 5-fold cross-validation on the z-scored training partition and aggregate per-fold accuracy as mean ± standard deviation. The **selected** k under a scheme is the argmax of mean CV accuracy. Crucially, the selection *rule* (argmax accuracy) is identical across schemes—only the fold-assignment rule changes.

Two caveats on how we report this. First, the accuracy at the argmax-selected k slightly over-estimates true performance, because the maximum over a noisy grid is upward-biased (Cawley & Talbot, 2010); here the effect is negligible, as the stratified-blocked curve is nearly flat near the optimum (k=1: 0.7778,

k=3: 0.7777, k=5: 0.7770). Second, the ± we report is the standard deviation of accuracy across the five folds from a single seed; because the folds share training data this is an informal measure of dispersion, not a calibrated confidence interval (Bengio & Grandvalet, 2004).

2.5 Cross-validation schemes (how the splitting actually works)

We implement all three schemes ourselves rather than delegating to a library, so we can state exactly how each one partitions the $N = 11{,}917$ training samples into 5 folds. In every scheme each sample appears in exactly one test fold, and a fold's training set is the complement of its test set.

Scheme A — shuffled 5-fold (leaky baseline). We generate an index array $[0, 1, \ldots, N-1]$ and apply a single seed-fixed Fisher–Yates permutation (NumPy `default_rng(42)`). The shuffled indices are cut into 5 contiguous equal blocks (the last block absorbs the remainder); block i is the test set of fold i and the other four blocks are its training set. Because the permutation is uniform, each test fold is a random 20% subset spread across the whole recording—so the immediate temporal neighbours of almost every test sample land in the training folds. This is the standard "`KFold(shuffle=True)`" protocol, and the leakiest possible scheme on a continuous time series.

Scheme B — naive blocked 5-fold. We take the time-ordered index array *without* shuffling and cut it into 5 contiguous chunks. Fold i's test set is chunk i (a single contiguous span of time); its training set is the remaining time. This eliminates lag-≤ 1 leakage but introduces a new problem: because the label runs are long and unevenly distributed, individual chunks can be near class-pure (e.g. one chunk at ~20% class-0 vs. the overall ~54%). Per-fold accuracy variance therefore explodes and the mean can fall *below* the majority baseline. We include it to make the failure mode visible—it is *not* a scheme we recommend.

Scheme C — stratified blocked 5-fold (honest). This is the protocol we recommend. We cut the time-ordered training partition into $M = 100$ short contiguous segments of ~119 samples (~0.93 s) each, preserving within-segment temporal continuity. We label each segment by its class-1 proportion, sort the segments by that proportion, and walk the sorted list in chunks of 5; within each chunk the five segments are assigned to the five folds by a seed-fixed random permutation. The effect is that each fold receives one segment from each quintile of the class-1-proportion distribution, so every fold's class balance is within ~3 pp of the overall ~54% class-0 balance. This keeps the temporal-isolation guarantee of blocked CV (neighbours stay together inside a segment) *and* restores the class-balance guarantee of stratified CV. One residual leak remains: we do not insert a seam gap *between* adjacent segments, so the two samples straddling each of the ~99 segment boundaries can land in different folds. This affects under 1% of training samples (99/11,917), so it can inflate the stratified-blocked accuracy by at most a point or two—a caveat we revisit when reconciling the CV and holdout estimates (§3.5). Figure 4 shows the chronological training partition with the naive-blocked 5-fold boundaries and the eye-state label strip, making the long, unevenly distributed label runs—the reason naive blocking fails and stratification is needed—visible at a glance.

2.6 Comparison models (sanity checks)

To test whether the leakage gap is KNN-specific or a generic property of this dataset, we run the same model-selection idea on three other classifiers from the palette. **LDA** has no tunable hyperparameter—one accuracy per scheme. **PCA→LDA** sweeps $n_{\text{comp}} \in \{1, 2, 3, 5, 7, 10\}$ and selects the n maximising mean stratified-blocked CV accuracy. **PCR-as-classifier** (ordinary least squares on the first n principal components, thresholded at 0.5) sweeps the same grid. These are supporting evidence, not the primary result.

2.7 Metrics

We report **accuracy** = (TP+TN)/N for both CV and holdout, and—because the chronological holdout is heavily class-imbalanced (90.6% class-0, as the final segment of the recording is a long eyes-open run)—we also report **sensitivity** (recall for closed= 1), **specificity** (recall for open= 0), and **balanced accuracy** (the mean of the two) on the holdout, since raw accuracy on a 91/9 split is misleading. We distinguish two

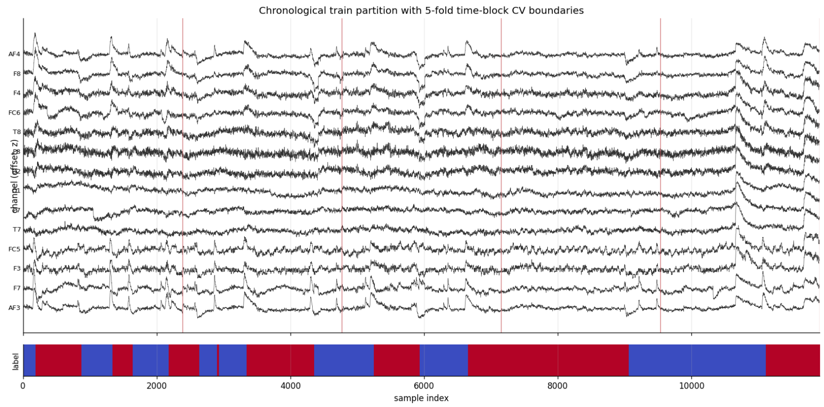

Figure 4: The chronological training partition (z-scored 14 channels, offset for readability) with the four naive-blocked 5-fold boundaries (red vertical lines) and the eye-state label strip (blue = eyes open, 0; red = eyes closed, 1). The long, unevenly distributed label runs mean the five contiguous naive-blocked folds are far from class-balanced, motivating the stratified-blocked scheme (Scheme C), which instead redistributes 100 short contiguous segments across folds.

Table 1: Per-scheme selected k and accuracy from the KNN k-sweep. All three schemes select $k = 1$; the reported accuracies span 47 percentage points.

CV scheme	Selected k	Mean accuracy ± SD
Shuffled (leaky baseline)	1	0.9728 ± 0.0037 ($\approx 97.3\%$)
Naive blocked	1	0.5004 ± 0.1068 ($\approx 50.0\%$)
Stratified blocked (honest)	**1**	$\mathbf{0.7778 \pm 0.0274}$ ($\approx 77.8\%$)

kinds of estimate throughout: the *within-recording* CV accuracy (resampled from the training partition) and the *out-of-time* holdout accuracy (a genuinely future, unseen segment); §3.5 shows they disagree, and we discuss why.

3 Results

3.1 Model selection: KNN under three CV schemes

Figure 5 is the headline figure. It plots KNN mean accuracy ± standard deviation as a function of k (log scale) under all three CV schemes on the same axes.

Three observations are immediate. The shuffled curve sits near 0.97 across all k, decaying roughly linearly on the log axis from 0.973 at $k = 1$ to ∼0.84 at $k = 201$—"nearly perfect" everywhere. The naive blocked curve sits near 0.50 across all k, at or below the majority baseline, with very large per-fold standard deviation (∼10–13 pp); this is fold construction failing, not the model failing (§4.3). The stratified blocked curve peaks at 0.778 at $k = 1$ and decays gracefully to ∼0.71 by $k = 201$, with a much smaller standard deviation (∼2.7–4.0 pp).

Applying the argmax-accuracy rule to each scheme yields Table 1.

The remarkable result is that **all three schemes select the same hyperparameter** ($k = 1$): the model-selection procedure is robust, but the *answer* it reports is not. The selected accuracy ranges from 50.0% (naive blocked) to 97.3% (shuffled)—a 47 *pp* spread—despite the procedure, the model, the data, and the selected k being identical.

3.2 Complexity, flexibility, and overfitting

For KNN, model flexibility is inversely related to k: $k = 1$ is the most flexible (a jagged, high-variance decision boundary that can fit any training point), and large k is more rigid (a smooth, high-bias boundary). The shape of the curves in Figure 5 is itself diagnostic of overfitting-via-leakage. Under *honest* stratified

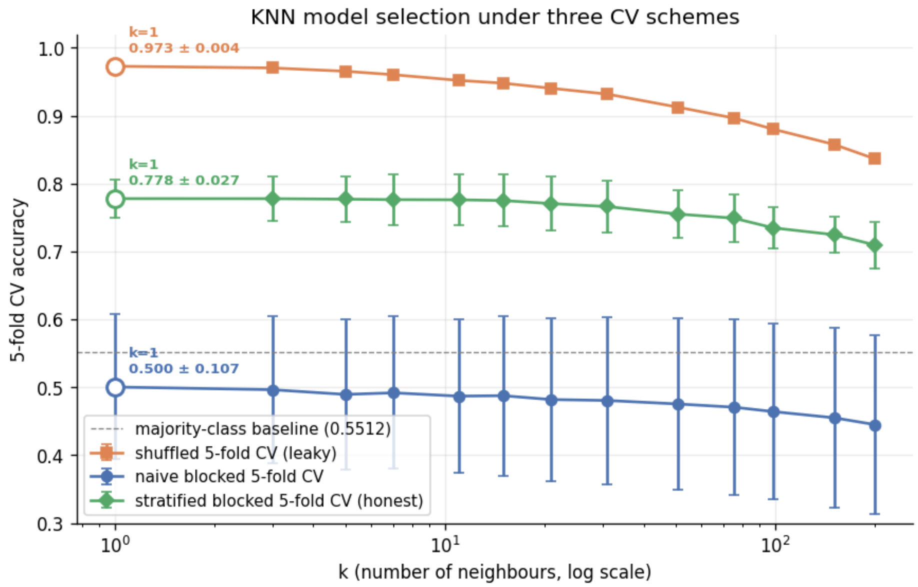

Figure 5: KNN model selection under three CV schemes (5-fold each; error bars are ± 1 SD across folds). Orange squares: shuffled CV (leaky baseline). Blue circles: naive blocked CV. Green diamonds: stratified blocked CV (honest). The horizontal dashed line is the majority-class baseline (0.5512). The open marker on each curve highlights the argmax-accuracy k and the corresponding accuracy.

blocked CV, accuracy *decreases* monotonically as k grows from 1 to 201 ($0.778 \to 0.710$): the flexible model generalises best, because the signal it exploits (short-range temporal smoothness within a segment) is real and local. Under *shuffled* CV the curve is pathologically flat-and-high ($\sim$0.97 everywhere): the model memorises the temporal twin of each test sample regardless of k. That flat-near-1 shape is the fingerprint of leakage-driven overfitting—the estimator looks like it cannot overfit because the "test" samples are near-duplicates of training samples.

3.3 Leakage-gap analysis

The methodological centrepiece is the **leakage gap**: the difference between an estimator's leaky shuffled-CV accuracy and its honest stratified-blocked-CV accuracy at the same selected hyperparameter. For KNN at $k = 1$,

$$\text{leakage gap} = 0.9728 - 0.7778 = 0.1950 \quad (\approx 19.5\,pp).$$

We attribute this gap to leakage of adjacent samples across the train/test fold boundary, not to model quality: the model is 1-NN in both cases and the training partition is identical—only the fold-assignment rule changes. Under shuffled CV the temporal neighbour of every test sample lands in a training fold with $\sim$80% probability and carries the same label with $\sim$99.7% probability (the lag-1 *label* autocorrelation); under stratified blocked CV that neighbour is reserved into the test sample's own segment in all but the <1% of cases that straddle a segment boundary. The honest within-recording accuracy at $k = 1$, $77.8\% \pm 2.7\%$, is $\sim$23 pp above the 55.12% baseline, so the shuffled estimate is clearly inflated (hypothesis part (ii)). Whether this within-recording signal *generalises out-of-time* is a separate question that the holdout answers less favourably (§3.5).

3.4 Supporting evidence: alternative classifiers

To rule out a generic explanation, we ran the same idea on three other classifiers (Table 2, Figure 6).

The pattern is exactly what we expect if the gap is caused by per-sample autocorrelation. LDA's decision rule is a single global linear projection of the 14-D space onto a 1-D axis; it cannot exploit per-sample similarity, so its gap is small (5.6 *pp*). PCA→LDA and PCR both project to a low-dimensional subspace before deciding, destroying most of the high-frequency sample-to-sample similarity KNN relies

Table 2: Per-model leakage gaps (shuffled CV minus stratified-blocked CV) at the selected hyperparameter. KNN's gap is the outlier: ∼4–8× larger than any other classifier in the palette.

Model	Selected hyperparam.	Shuffled CV	Strat. blocked CV	Leakage gap
LDA	— (no sweep)	0.6471 ± 0.0063	0.5911 ± 0.0474	$+5.6\,pp$
PCA→LDA	$n = 3$	0.5574 ± 0.0147	0.5339 ± 0.0438	$+2.4\,pp$
PCR (classifier)	$n = 2$	0.5528 ± 0.0087	0.5250 ± 0.0176	$+2.8\,pp$
KNN	$\mathbf{k = 1}$	$\mathbf{0.9728 \pm 0.0037}$	$\mathbf{0.7778 \pm 0.0274}$	$\mathbf{+19.5\,pp}$

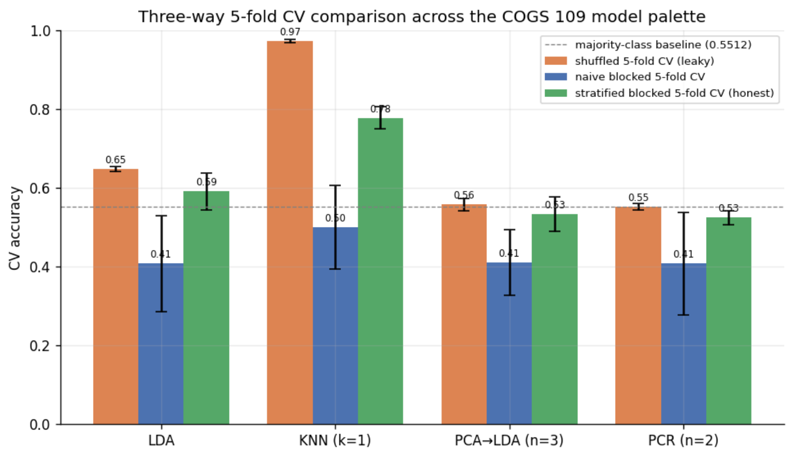

Figure 6: Three-way 5-fold CV accuracy across the four palette classifiers. Orange: shuffled; blue: naive blocked; green: stratified blocked. The KNN bar group has by far the largest shuffled-to-stratified-blocked gap, consistent with KNN being uniquely vulnerable to lag-1 autocorrelation.

on; their gaps are smaller still (2.4 and 2.8 pp). KNN at $k = 1$ is the only classifier whose prediction depends on the identity of the single nearest training sample—and its gap is 4–8× larger than any other.

3.5 Model estimation: final parameters and accuracy

Final model and "parameters". The selected model is KNN with $k = 1$ on the z-scored 14-channel features. KNN is non-parametric, so the "fitted parameters" are not coefficients but (i) the selected hyperparameter $k = 1$ and (ii) the stored training set together with the per-channel z-scoring constants (μ_c, σ_c). Following the standard convention of refitting the final model on all available data once the hyperparameter is fixed (James et al., 2021), the deployable model recomputes (μ_c, σ_c) and sets the 1-NN store on the *entire* cleaned dataset ($n = 14{,}976$); cross-validation is used only for selection, not for the final fit. For a concrete parametric contrast, the LDA comparison model *does* have estimable coefficients (`figures/10_lda_coefficients.png`): the largest-magnitude discriminant weights fall on the temporal (T7, T8) and frontal (F8, F7) channels, with parietal P7 also prominent, while the occipital channels O1/O2 are small-to-negligible. We therefore do *not* attribute the discriminant to posterior alpha rhythms; large temporal and frontal weights are more consistent with eye-movement and muscle (EMG) artefact and with the per-sensor amplitude-range differences that Roesler & Suendermann (2013) identify as discriminative on this dataset. A linear discriminant on raw per-sample voltages cannot in any case isolate band-limited alpha power.

Final prediction accuracy. The headline estimate is the honest *within-recording* cross-validated accuracy of the selected model ($k = 1$, stratified blocked CV): **77.8%** $\pm$ **2.7%**, versus the 55.12% majority baseline.

Reconciling the holdout: out-of-time generalisation is weaker. The 20% chronological holdout is the single most leakage-resistant test in the study—it is genuinely future data, separated from training by

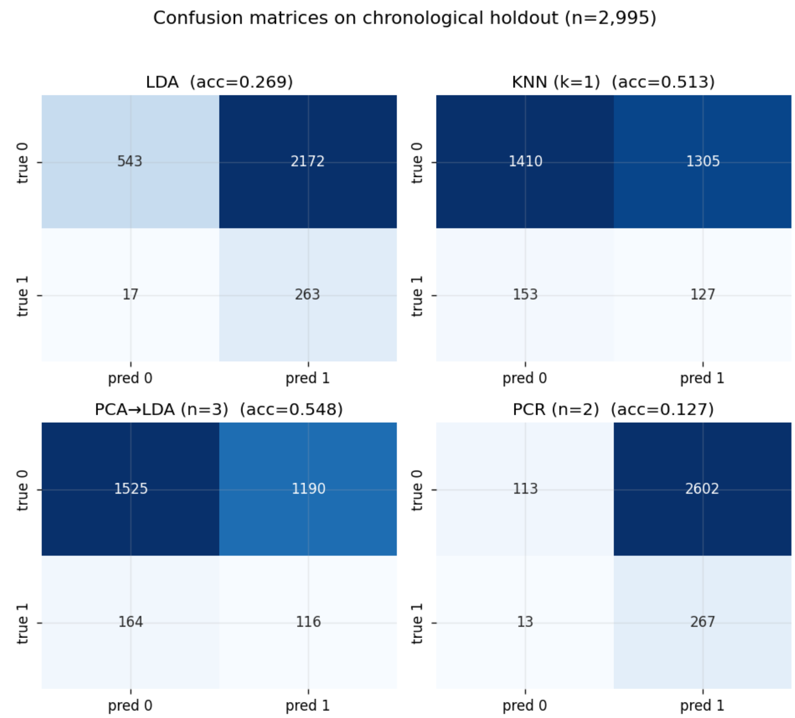

Figure 7: Per-model confusion matrices on the 20% chronological holdout test set (rows: true; columns: predicted). The holdout is dominated by eyes-open samples (~91% class-0), so accuracy here is mostly driven by specificity. The number in each panel is that model's holdout accuracy.

the 64-sample seam gap—so we confront it directly rather than discarding it. On the holdout, KNN ($k = 1$) scores only 51.3% accuracy with sensitivity 0.45 and specificity 0.52, i.e. a balanced accuracy of ≈ 0.49: essentially chance, and far below the holdout's own 90.6% majority baseline (the final segment is a long eyes-open run, so the holdout is 90.6% class-0 versus 46.5% in training; Figure 7). The within-recording CV cannot see this prior shift because every fold is rebalanced to the training distribution, whereas the holdout can. The honest reading is therefore split: the leakage result (part (ii)) is robust, but the claim that the model captures a *generalisable* eye-state signal (part (i)) holds only *within* the recording and is *not* corroborated out-of-time on this single ~117 s session.

4 Conclusions and Discussion

4.1 What we conclude about the hypothesis

Our hypothesis holds in part. Part (ii) is strongly supported: the shuffled-CV estimate (97.3%) is inflated by 19.5 *pp* relative to the honest stratified-blocked estimate, and that inflation is leakage attributable purely to the CV scheme—the model, the selected k, and the training data are identical across schemes (§3.3). Part (i) is supported only within the recording: the honest *within-recording* KNN accuracy (77.8%) beats the 55.12% baseline by ~23 pp, but the out-of-time holdout (balanced accuracy ≈ 0.49, §3.5) does *not* corroborate a generalisable voltage-to-state rule. We report this candidly: on a single-subject, single-session recording, within-recording resampling and genuine out-of-time prediction disagree, and only the leakage finding is unambiguous.

> *Shuffled CV says we have a 97% classifier. Honest stratified blocked CV says we have a 78% classifier. The difference is leakage, not signal.*

4.2 Why KNN is uniquely vulnerable

The mechanism is the one quantified in §3.3: because the per-channel lag-1 voltage autocorrelation is ≈ 0.97, a test sample's nearest neighbour is almost always its temporal twin, whose label matches with probability ≈ 0.997. Shuffled CV leaves that twin in a training fold, so 1-NN exceeds 95% by memorisation; stratified blocked CV reserves it into the test sample's own segment, dropping accuracy to 77.8%. Models that average over many samples (LDA, PCA→LDA, PCR) cannot exploit per-sample twins and show only 2–6 *pp* gaps.

4.3 Why naive blocked CV underperforms the baseline

The five contiguous chunks are not class-balanced: with only 23 label transitions in the recording, several chunks straddle few or zero transitions, and per-chunk class proportions stray far from the global 55/45 split. A classifier trained on four mostly-open chunks then predicts "open" on a held-out mostly-closed chunk, scoring *below* the majority baseline. This is a fold-stratification failure, not a model failure; stratifying contiguous segments (Scheme C) fixes it, which is why stratified blocked CV recovers ∼28 pp over naive blocked.

4.4 Implications and future work

A non-trivial fraction of tutorials and student projects on this dataset report 95–98% KNN accuracy under shuffled k-fold. Our analysis suggests those numbers are a CV-protocol artefact: the honest within-recording accuracy is 77.8% and the extra ∼20 pp is adjacent-sample leakage. Tellingly, even the dataset's originators obtain their best results with instance-based learners (KStar/IB1, $\approx 97\%$) under shuffled 10-fold CV while non-instance-based methods score far worse (Roesler & Suendermann, 2013)—exactly the signature of leakage that instance-based models are uniquely able to exploit. This echoes broader cautions about i.i.d. CV on autocorrelated signals (Bergmeir & Benítez, 2012). The practical recommendation is to use blocked, class-stratified CV (or an explicit out-of-time holdout with a seam gap) whenever samples are temporally autocorrelated. Limitations of our study: it is a single subject on consumer hardware over ∼117 s, and we use per-sample voltages with no windowed features. Future work should (i) add windowed features (rolling means/variances, lagged values), which would give KNN more meaningful neighbours and likely shrink the leakage gap; (ii) test cross-subject generalisation on a multi-subject corpus; and (iii) extend the k sweep to characterise the leakage curve more finely. The project is, above all, a worked example of how a methodologically sound model-selection procedure can still be misled by an upstream choice—the CV scheme—unrelated to the model itself.

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